Introduction to Software Metrics and Measurement

Davide Falessi

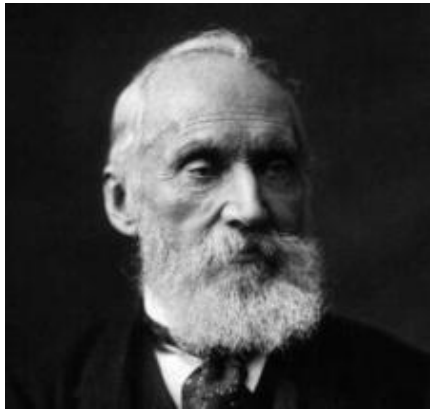
Agenda

- Introduction
- Measures, Metrics and Measurements
- Types of Scales

Importance of Measurement

When you cannot measure it […] you have scarcely in your thoughts advanced to the state of Science, whatever the matter may be.

If you can not measure it, you can not improve it.



(William Thomson,) Lord Kelvin

Belfast, 26.06.1824 – Largs, 17.12.1907

Physicist, and Engineer

Importance of Software Measures



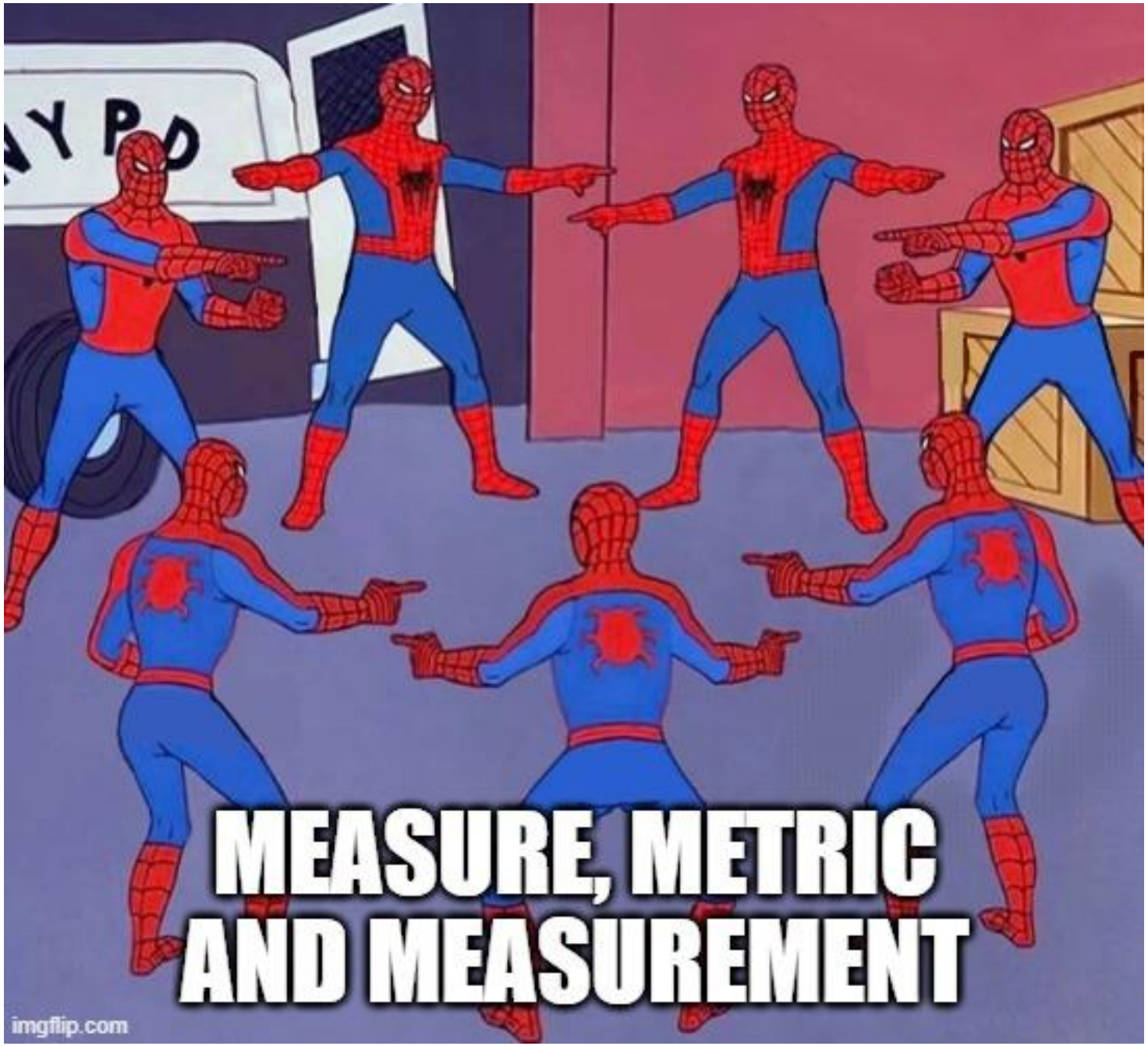
Tom DeMarco is an author, teacher, and speaker on Software engineering topics. He received the1986 **Warnier Prize** and the 1999 **Stevens Award**.".

You can't control what you can't measure.

Tom DeMarco

Hazleton, Pennsylvania, 20 August 1940

- *Controlling Software Projects: Management, Measurement, and Estimation* (Prentice Hall/Yourdon Press, 1982).


MEASURE, METRIC
AND MEASUREMENT
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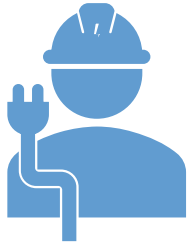
Metric

- A metric is a scale or system of measurement used to quantify a measure
  - A metric defines the rules for assigning numbers to measures
    - Examples: lines of code (LOC), cyclomatic complexity (CC), defect density (DD)

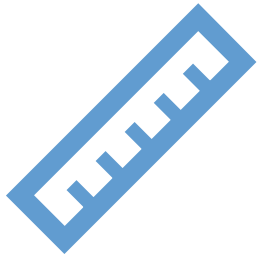
Measurement



Measurement is the process of assigning numbers to measures using a metric.



Measurement is essential for objective evaluation, improvement, and decision-making in software engineering.



A measurement can be made on a single instance or a collection of instances of a measure.

Measure

- A measure is a numerical representation of a characteristic or attribute of a software product or process
  - Examples: 30 MM

Example

- ISPW project = 5000 LOC
  - Measure, Metric, Measurement?

Aims of software measures



Evaluation: Software measures can be used to evaluate the quality and effectiveness of software development processes, software products, and software maintenance activities.



Improvement: By measuring various aspects of software development, it is possible to identify areas for improvement and implement changes that can lead to better software quality, productivity, and efficiency.



Planning: Software measures can be used to estimate the resources required for software development projects, such as time, effort, and cost. They can also be used to plan and schedule project tasks and milestones.



Control: Software measures can be used to monitor and control the progress of software development projects, ensuring that they are on track and within budget.



Communication: Software measures can be used to communicate the status and quality of software development projects to stakeholders, such as project managers, customers, and other members of the development team.

Properties of good metrics

Relevant: The metric should measure a meaningful and important attribute of the software product or process

Objective: The metric should be based on verifiable and reproducible data, not subjective opinions or estimates

Reliable: The metric should produce consistent results under different conditions and by different evaluators

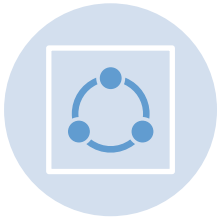
Consistent: The metric should be consistent with the goals and context of the software project

Easy to use: The metric should be easy to understand, compute, and interpret, and not require excessive effort or resources

Examples of Metrics



Software size: LOC, function points (FP), object points (OP)



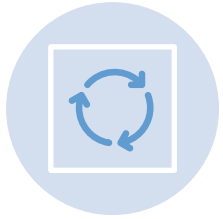
Software complexity: CC, information flow metrics (IFM), Halstead's software science metrics



Software quality: defect density (DD), fault frequency (FF), reliability



Software effort: cost, effort, schedule



Software process: productivity, efficiency, cycle time, lead time

Types of Scales in Software Measurement and Analysis

Introduction to scales



Software measurement and analysis involve using quantitative metrics to evaluate and improve software quality, productivity, and efficiency.



To make meaningful measurements, we must use appropriate scales that reflect the data's nature.



This presentation will discuss the scales commonly used in software measurement and analysis.


NOMINAL
ORDINAL
INTERVAL
RATIO
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Nominal

- Nominal scale is the simplest type of scale, where values are assigned to categories or labels without any order or magnitude.
- Nominal scales represent qualitative data, such as software defects by type or severity.
- Examples: Defect Types: UI, Logic, Performance, Security

Ordinal

- Ordinal scale is a scale where values are assigned to categories or labels in ranked order, but the magnitude of the differences between values is not meaningful.
- Ordinal scales represent data that can be ordered but not measured, such as software defects by priority.
- Examples: Defect Priority: Critical, High, Medium, Low

Interval

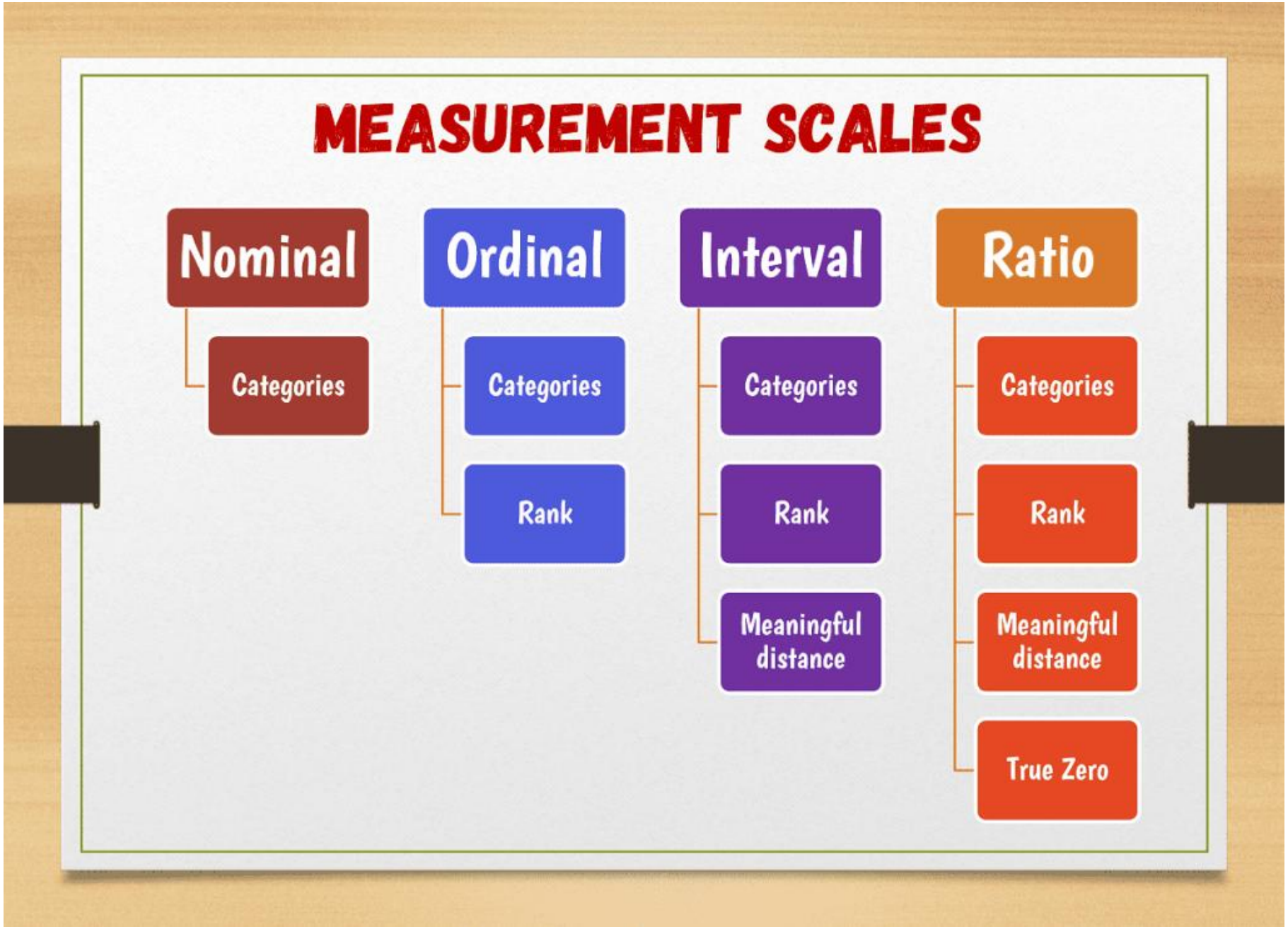
- Interval scale is a type of scale where values are assigned to categories or labels in ranked order, and the magnitude of the differences between values is meaningful.
- However, there is no true zero point on the scale.
- Interval scales are used to represent data that can be measured and compared, such as software development effort in hours or days.
- Examples: Effort: 1 hour, 2 hours, 3 hours, etc.

Ratio

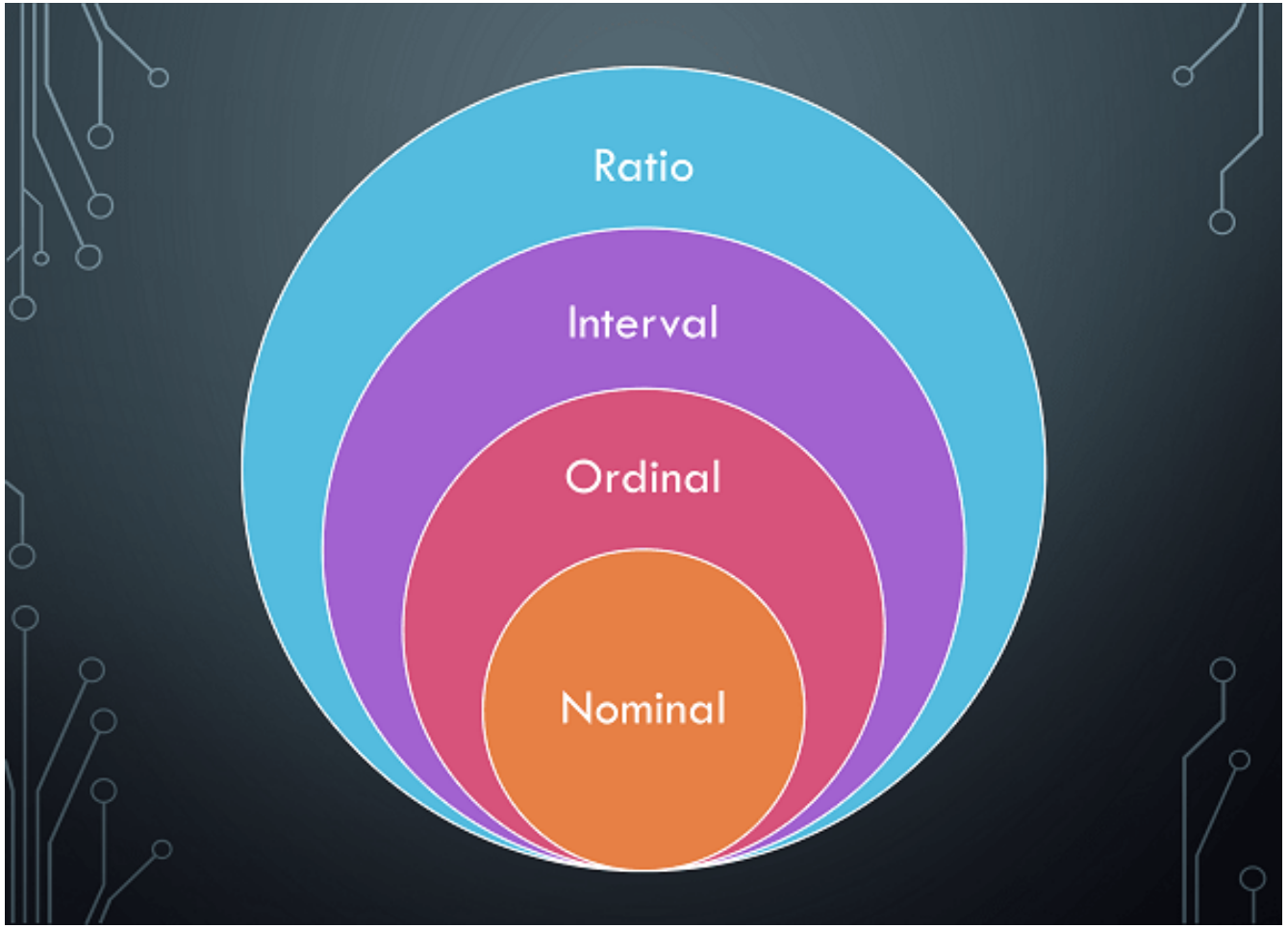
- Ratio scale is the most sophisticated type of scale, where values are assigned to categories or labels in ranked order, the magnitude of the differences between values is meaningful, and there is a true zero point on the scale.

- Ratio scales are used to represent data that can be measured and compared, such as lines of code, defects per KLOC, or test coverage.

- Examples: Effort: 1 hour, 2 hours, 3 hours, etc.


MEASUREMENT SCALES
Nominal
Categories
Ordinal
Categories
Rank
Interval
Categories
Rank
Meaningful distance
Ratio
Categories
Rank
Meaningful distance
True Zero

More on scale



Statistics for Scales

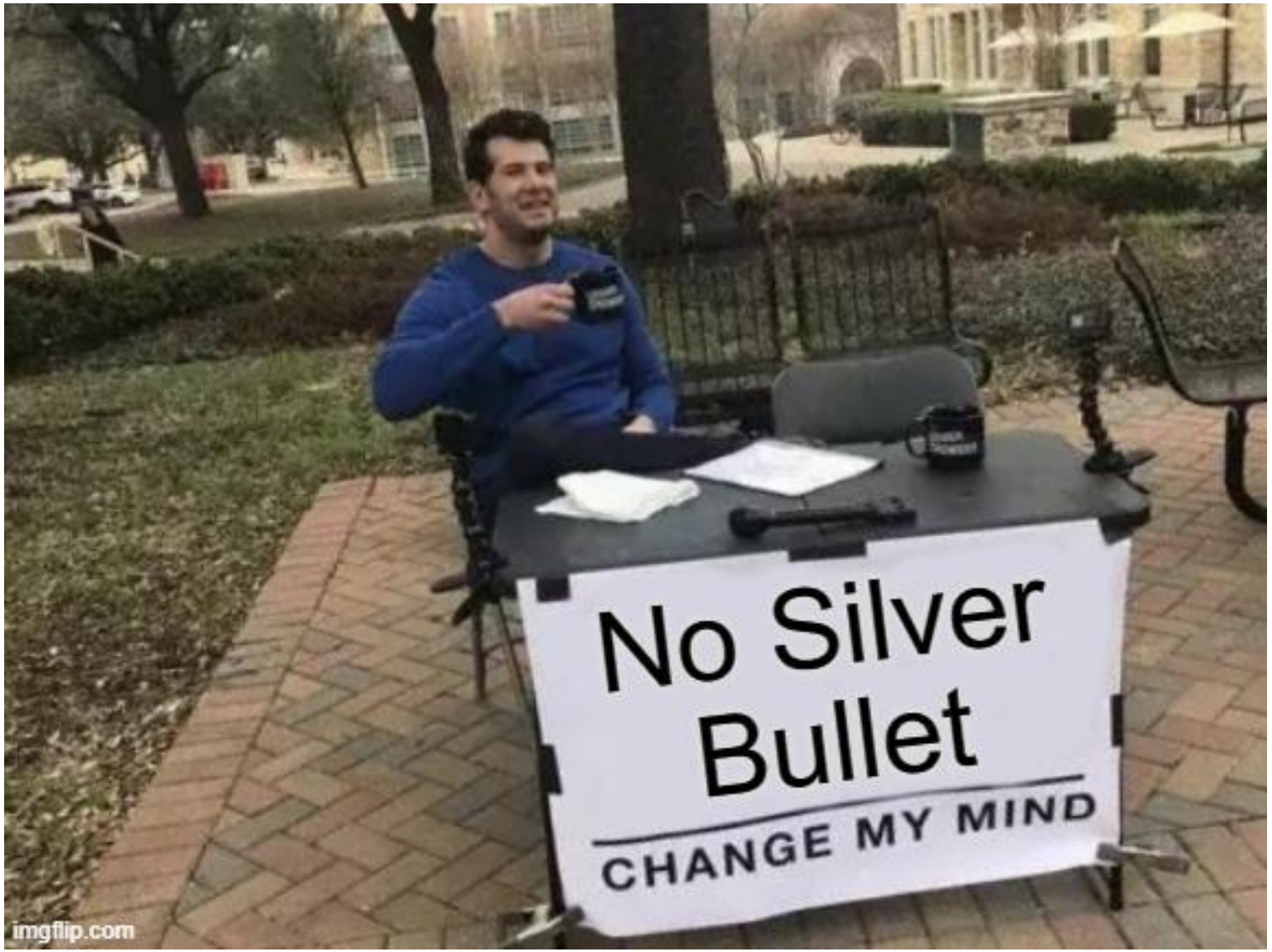
Statistic / *Scale type*	Measure of: Central tendency	Dispersion	Dependency
Nominal	Mode	No., Frequency	
Ordinal	Median	Interval of variation	Spearman Correlation
Interval	Mean	Std. Deviation, Variance, Range	Pearson Correlation
Ratio	Geometric mean	Coefficient of variation	Statistical Tests

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Examples

- Book chapters?
- Colors?
- Gates?
- Train or bus ID?
- Grades?
- Usability?
- Security?
- Profit?

Scales



Conclusions on scales

- The choice of scale used in software measurement and analysis depends on the nature of the data being measured and the intended use of the results.
- There are silver bullets.
- By understanding the different types of scale, we can ensure that our measurements are appropriate, accurate, and meaningful, leading to better decision-making and improved software quality.

References

- Fenton N, Bieman J. Software metrics: a rigorous and practical approach. CRC press; 2014 Oct 1.